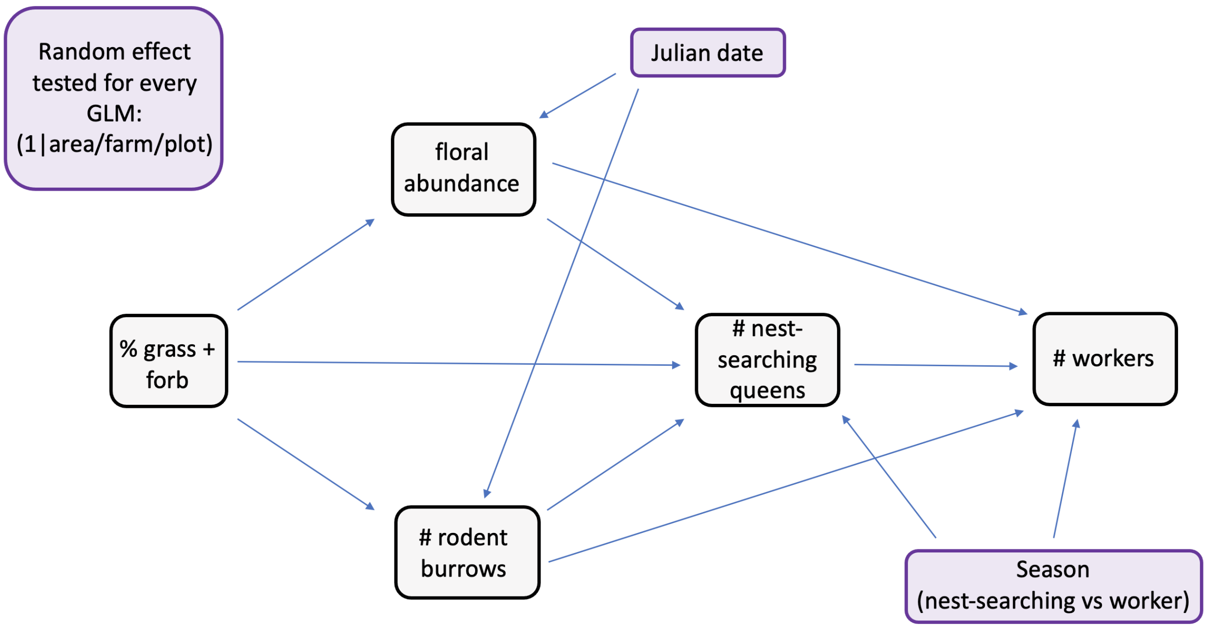
ch1_glm_clean

Predicted path diagram



Load and clean data

```r
#load packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(DHARMa)
```

```
## This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')
```

```r
library(ordinal)
```

```
##
## Attaching package: 'ordinal'
##
## The following object is masked from 'package:dplyr':
##
##     slice
```

```r
library(MASS)
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
```

```r
library(glmmTMB)

#read in data
data <- read.csv("06_cleandata/combined_plot_data.csv")

#change character columns to factors
surveys <- data %>%
   mutate(
     site_id = as.factor(site_id),
     round = as.factor(round),
     plot_id = as.factor(plot_id),
     bee_date = as.Date(bee_date),
     bee_season_type = as.factor(bee_season_type),
     area = as.factor(area))

#scale continuous predictors
surveys$julian.date_scaled <- as.numeric(scale(surveys$julian.date))
surveys$bare_perc_scaled <- as.numeric(scale(surveys$bare_perc))
surveys$grass_perc_scaled <- as.numeric(scale(surveys$grass_perc))
surveys$forb_perc_scaled <- as.numeric(scale(surveys$forb_perc))
surveys$grass_forb_perc_scaled <- as.numeric(scale(surveys$grass_forb_perc))
surveys$grass_h_scaled <- as.numeric(scale(surveys$grass_h))
surveys$forb_h_scaled <- as.numeric(scale(surveys$forb_h))
surveys$floral_perc_scaled <- as.numeric(scale(surveys$floral_perc))
surveys$grass_forb_floral_perc_scaled <- as.numeric(scale(surveys$grass_forb_floral_perc))
surveys$bombus_floral_perc_scaled <- as.numeric(scale(surveys$bombus_floral_perc))
surveys$flood_perc_scaled <- as.numeric(scale(surveys$flood_perc))
surveys$dry_root_perc_scaled <- as.numeric(scale(surveys$dry_root_perc))
surveys$bryo_perc_scaled <- as.numeric(scale(surveys$bryo_perc))
```

```r
surveys$bare_flood_dry_bryo_perc_scaled <- as.numeric(scale(surveys$bare_flood_dry_bryo_perc))
surveys$floral_richness_scaled <- as.numeric(scale(surveys$floral_richness))
surveys$bombus_floral_richness_scaled <- as.numeric(scale(surveys$bombus_floral_richness))
surveys$bombus_floral_abun_scaled <- as.numeric(scale(surveys$bombus_floral_abun))
surveys$bombus_floral_abun_nv_scaled <- as.numeric(scale(surveys$bombus_floral_abun_nv))
surveys$nest_searching_queens_scaled <- as.numeric(scale(surveys$nest_searching_queens))
surveys$workers_scaled <- as.numeric(scale(surveys$workers))
surveys$forage_vaco_scaled <- as.numeric(scale(surveys$forage_vaco))
surveys$num_burrows_scaled <- as.numeric(scale(surveys$num_burrows))
surveys$num_active_burrows_scaled <- as.numeric(scale(surveys$num_active_burrows))
surveys$num_inactive_burrows_scaled <- as.numeric(scale(surveys$num_inactive_burrows))
surveys$num_active_dm_scaled <- as.numeric(scale(surveys$num_active_dm))
surveys$num_dm_size_scaled <- as.numeric(scale(surveys$num_dm_size))
surveys$num_inactive_dm_scaled <- as.numeric(scale(surveys$num_inactive_dm))
surveys$prop_inactive_burrows_scaled <- as.numeric(scale(surveys$prop_inactive_burrows))
surveys$num_runway_burrows_scaled <- as.numeric(scale(surveys$num_runway_burrows))

#filter to only include field data
field_data <- surveys %>%
  filter(plot_id != "d")
```

Floral abundance vs field management (%grass+forb)

Use cumulative link model aka ordinal logistic regression because bombus-relevant floral abundance has:

- discrete levels (0-5 scale) representing binned counts of blooms
- ordered categories (higher numbers represent more blooms)
- unequal intervals between categories

Steps

1) Tried running CLM with different random effect structures (full = 1|area/site/plot) then compared AIC values to determine best model.

- determined that model without random effect had best fit

```r
floral_full <- clmm(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
                         julian.date_scaled +
                         (1|area/site_id/plot_id),
                       link = "logit",
                       data = field_data)

floral_noArea <- clmm(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
                         julian.date_scaled +
                         (1|site_id/plot_id),
                       link = "logit",
                       data = field_data)

#floral_plotOnly <- clmm(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
```

```r
#                         julian.date_scaled +
 #                        (1|plot_id),
  #                     link = "logit",
   #                    data = field_data)
##can't use since there are only 2 levels

floral_siteOnly <- clmm(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
                          julian.date_scaled +
                          (1|site_id),
                        link = "logit",
                        data = field_data)

floral_AreaPlot <- clmm(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
                          julian.date_scaled +
                          (1|area/plot_id),
                        link = "logit",
                        data = field_data)
floral_noRE <- polr(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
                          julian.date_scaled,
                        method = "logistic",
                        data = field_data)

#compare models AIC
AIC(floral_full, floral_noArea, floral_siteOnly, floral_AreaPlot, floral_noRE)
```

```
##                  df      AIC
## floral_full       9 441.1757
## floral_noArea     8 439.1757
## floral_siteOnly   7 437.1757
## floral_AreaPlot   8 439.1757
## floral_noRE       6 435.1757
```

```r
##shows that model without random effect has best fit
```

2) Compare AIC between model with and without cover_type as a covariate.

- determined that model without cover type as covariate was better

```r
floral_noRE_cover <- polr(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
                             cover_type +
                          julian.date_scaled,
                        method = "logistic",
                        data = field_data)

AIC(floral_noRE_cover, floral_noRE)
```

```
##                    df      AIC
## floral_noRE_cover  7 437.1425
## floral_noRE        6 435.1757
```

```r
##not including cover is better
```

3) Run final model and look at output

```r
floral_noRE <- polr(as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
                         julian.date_scaled,
                       method = "logistic",
                       data = field_data)

summary(floral_noRE)
```

```
##
## Re-fitting to get Hessian

## Call:
## polr(formula = as.factor(bombus_floral_abun) ~ grass_forb_perc_scaled +
##     julian.date_scaled, data = field_data, method = "logistic")
##
## Coefficients:
##                           Value Std. Error  t value
## grass_forb_perc_scaled -0.01279     0.1825 -0.07006
## julian.date_scaled     -0.23080     0.1525 -1.51353
##
## Intercepts:
##     Value   Std. Error t value
## 0|1 -1.4085  0.2183    -6.4529
## 1|2 -0.9392  0.1939    -4.8422
## 2|3 -0.1953  0.1765    -1.1065
## 3|4  0.6093  0.1838     3.3149
##
## Residual Deviance: 423.1757
## AIC: 435.1757
```

Findings

No significant effect of either grass+ forb% or julian date.

Rodent burrows vs field management (%grass+forb)

Since the number of rodent burrows was count data, I knew I had to use either a poisson or negative binomial GLM family. I started with the simplest family (poisson) first.

Steps

1) Try different random effect structures in the poisson. Compare models using likelihood ratio tests to determine best random effect structure.

- best fit = 1|site_id/plot_id –> **area not included**

```r
burrows_full <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                             julian.date_scaled +
                             (1|area/site_id/plot_id),
                           family = poisson,
                           data = field_data)


burrows_noArea <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                             julian.date_scaled +
                             (1|site_id/plot_id),
                           family = poisson,
                           data = field_data)

burrows_plotOnly <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                             julian.date_scaled +
                             (1|plot_id),
                           family = poisson,
                           data = field_data)

burrows_siteOnly <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                             julian.date_scaled +
                             (1|site_id),
                           family = poisson,
                           data = field_data)

burrows_AreaPlot <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                             julian.date_scaled +
                             (1|area/plot_id),
                           family = poisson,
                           data = field_data)


#compare models using maximum likelihood test
anova(burrows_full, burrows_noArea, burrows_plotOnly, burrows_siteOnly, burrows_AreaPlot)
```

```
## Data: field_data
## Models:
## burrows_plotOnly: num_burrows ~ grass_forb_perc_scaled + julian.date_scaled + (1 | , zi=~0, disp=~1
## burrows_plotOnly:     plot_id), zi=~0, disp=~1
## burrows_siteOnly: num_burrows ~ grass_forb_perc_scaled + julian.date_scaled + (1 | , zi=~0, disp=~1
## burrows_siteOnly:     site_id), zi=~0, disp=~1
## burrows_noArea: num_burrows ~ grass_forb_perc_scaled + julian.date_scaled + (1 | , zi=~0, disp=~1
## burrows_noArea:     site_id/plot_id), zi=~0, disp=~1
## burrows_AreaPlot: num_burrows ~ grass_forb_perc_scaled + julian.date_scaled + (1 | , zi=~0, disp=~1
## burrows_AreaPlot:     area/plot_id), zi=~0, disp=~1
## burrows_full: num_burrows ~ grass_forb_perc_scaled + julian.date_scaled + (1 | , zi=~0, disp=~1
## burrows_full:     area/site_id/plot_id), zi=~0, disp=~1
##                  Df    AIC    BIC   logLik deviance  Chisq Chi Df Pr(>Chisq)
## burrows_plotOnly  4 4819.5 4831.3 -2405.75   4811.5
## burrows_siteOnly  4 1672.5 1684.3  -832.25   1664.5 3147.0      0     <2e-16
## burrows_noArea    5  621.0  635.7  -305.51    611.0 1053.5      1     <2e-16
## burrows_AreaPlot  5 2257.3 2272.0 -1123.66   2247.3    0.0      0          1
## burrows_full      6  622.3  640.0  -305.16    610.3 1637.0      1     <2e-16
```

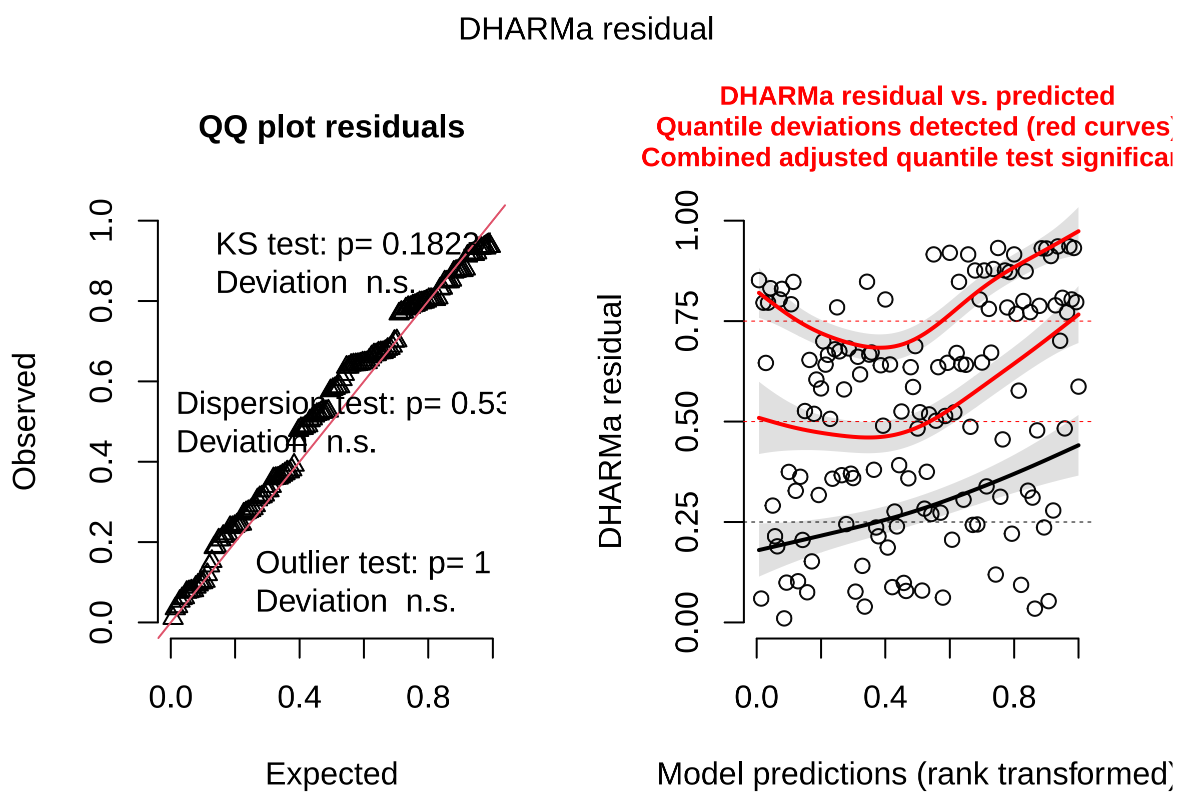
```
##
## burrows_plotOnly
## burrows_siteOnly ***
## burrows_noArea   ***
## burrows_AreaPlot
## burrows_full     ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##results tells us the burrows_noArea is best model (lowest AIC)
```

2) Check diagnostics of the model using DHARMa package.

- Combined adjusted quantile test significant

```
#simulate residuals
residuals_burrows_noArea <- simulateResiduals(fittedModel = burrows_noArea, plot = TRUE)
```



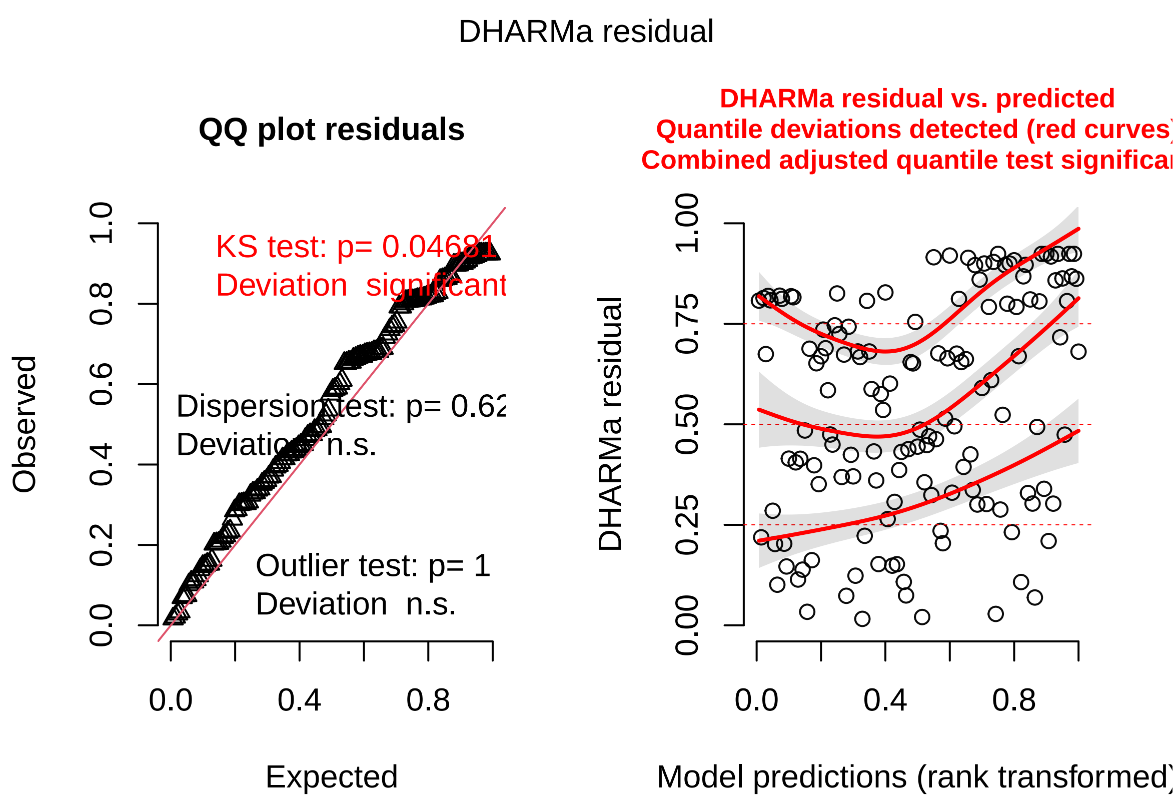
3) Try negative binomial instead and check diagnostics.

- Combined adjusted quantile test still significant.

```r
burrows_noArea_negBin <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                           julian.date_scaled +
                           (1|site_id/plot_id),
                         family = nbinom1, #use nbinom1 since nbinom2 doesn't converge
                         data = field_data)

#simulate residuals
residuals_burrows_noArea_negBin <- simulateResiduals(fittedModel = burrows_noArea_negBin, plot = TRUE)
```

```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```

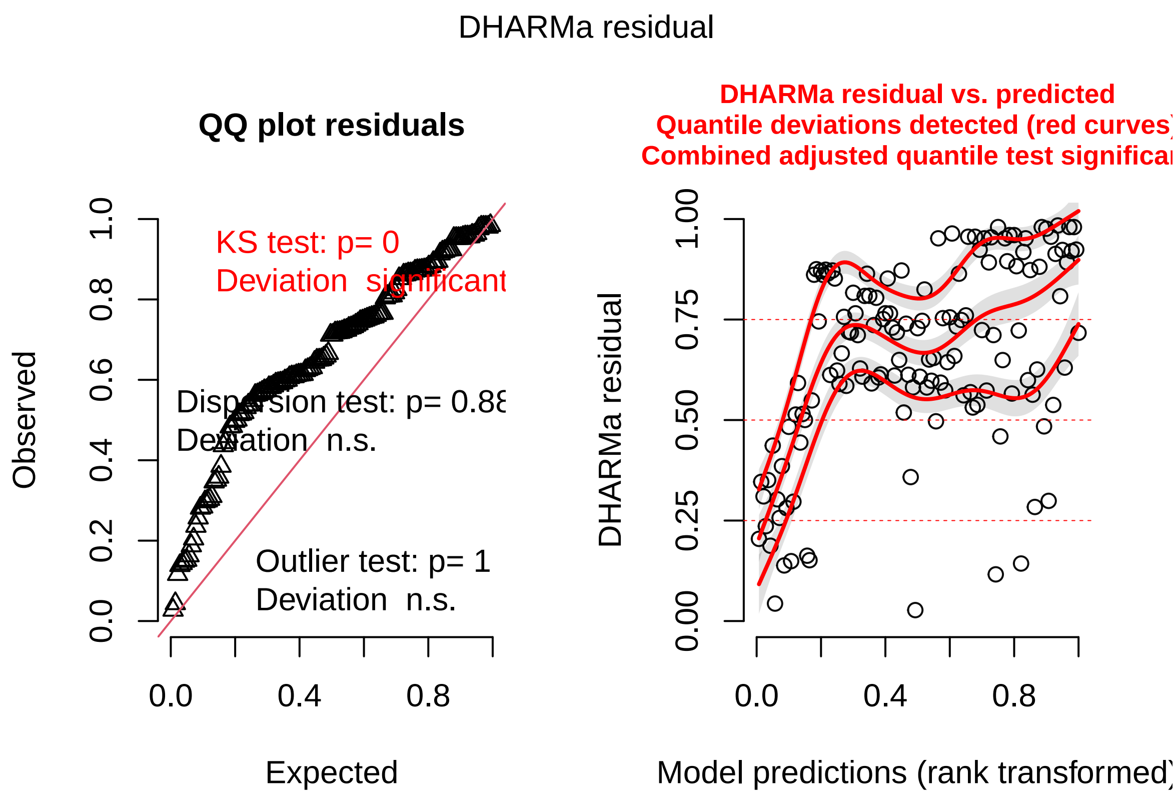


4) Try zero-inflated poisson and check diagnostics.

- KS test and combined adjusted quantile test still significant.

```r
burrows_zi_pois <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                           julian.date_scaled +
                           (1|site_id/plot_id),
                         ziformula = ~ grass_forb_perc_scaled +
                           julian.date_scaled +
                           (1|site_id/plot_id), # zero-inflation model
                         family = poisson,
                         data = field_data)
```

```r
#simulate residuals
residuals_burrows_zi <- simulateResiduals(fittedModel = burrows_zi_pois, plot = TRUE)
```

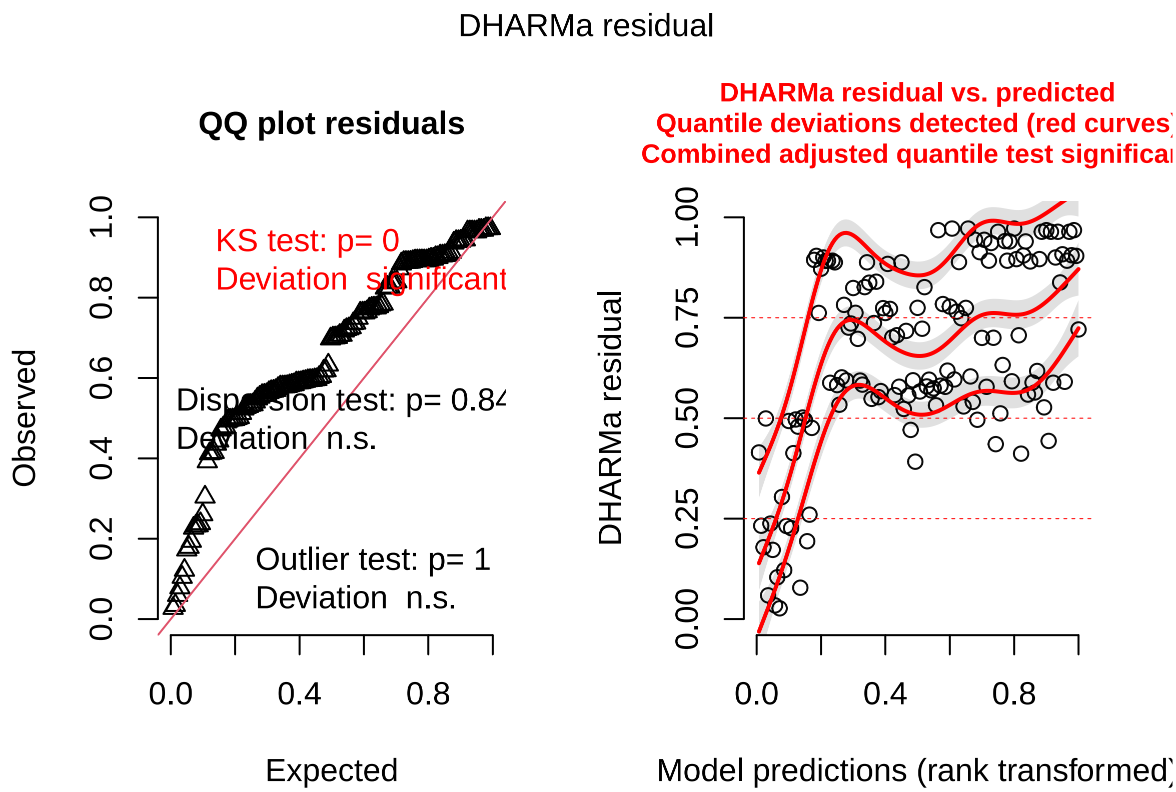


5) Try zero-inflated negative binomial and check diagnostics.

- KS test and combined adjusted quantile test still significant.

```r
burrows_zi_nbin <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                             julian.date_scaled +
                             (1|site_id/plot_id),
                           ziformula = ~ grass_forb_perc_scaled +
                             julian.date_scaled +
                             (1|site_id/plot_id), # zero-inflation model
                           family = nbinom2,
                           data = field_data)

#simulate residuals
residuals_burrows_zi_nbin <- simulateResiduals(fittedModel = burrows_zi_nbin, plot = TRUE)
```



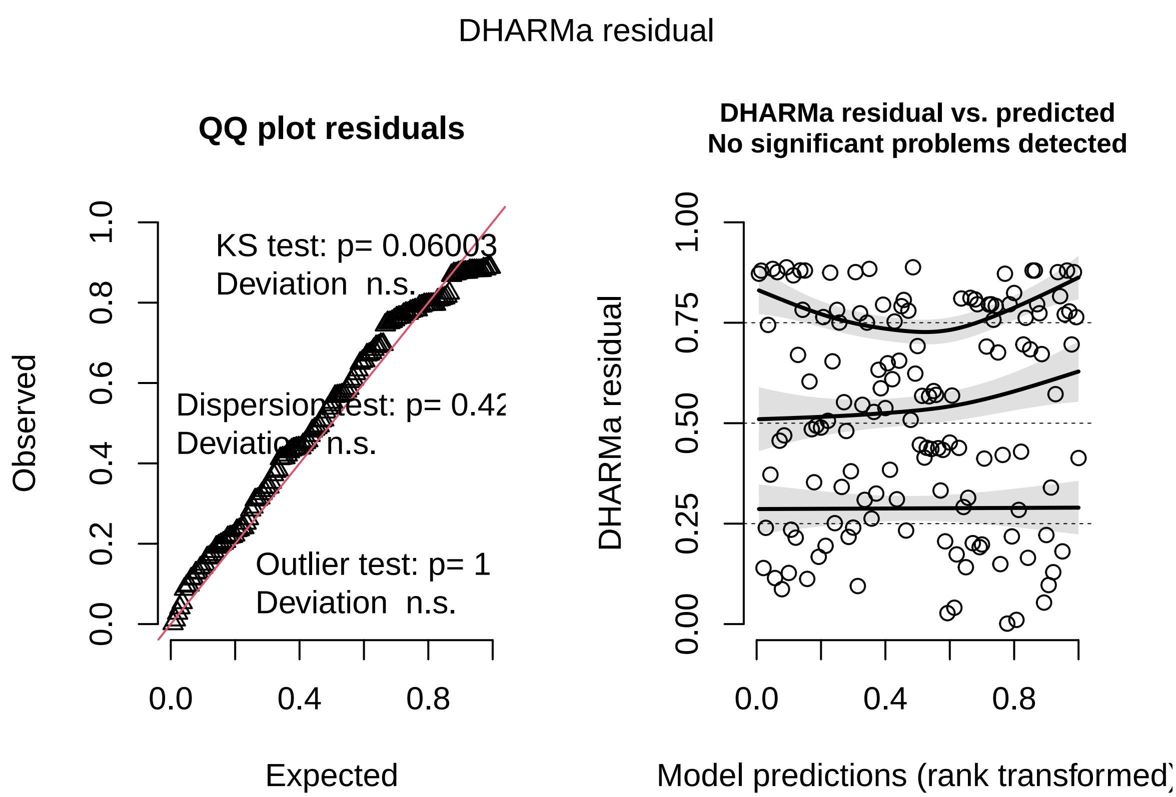

```
#signficant KS test and combined adjusted quantile test
```

6) Stick with poisson but try adding cover_type as an additional covariate and check diagnostics.

- none of the tests were significant! –> good to keep this one

```
burrows_pois_cover <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
                                          julian.date_scaled +
                                          cover_type +
                              (1|site_id/plot_id),
                            family = poisson,
                            data = field_data)

#simulate residuals
residuals_burrows_pois_cover <- simulateResiduals(fittedModel = burrows_pois_cover, plot = TRUE)
```




```
summary(burrows_pois_cover)
```

```
##  Family: poisson  ( log )
## Formula:
## num_burrows ~ grass_forb_perc_scaled + julian.date_scaled + cover_type +
##     (1 | site_id/plot_id)
## Data: field_data
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    621.1     638.7    -304.5     609.1       134
##
## Random effects:
##
## Conditional model:
##  Groups          Name        Variance Std.Dev.
##  plot_id:site_id (Intercept) 1.254    1.120
##  site_id         (Intercept) 3.554    1.885
## Number of obs: 140, groups:  plot_id:site_id, 24; site_id, 12
##
## Conditional model:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)            0.5012469  0.8700010   0.576    0.565
## grass_forb_perc_scaled 0.0008310  0.0446708   0.019    0.985
## julian.date_scaled     0.0002004  0.0184457   0.011    0.991
## cover_typegrass        1.7447169  1.2077682   1.445    0.149
```

Findings

Neither grass + forb % nor julian date had significant effects on rodent burrow abundance. However, cover type showed a marginally insignificant trend (p=0.149), where grassy sites had a higher abundance of burrows.

Number of nest-searching queens vs flowers, rodents, and field management technique

Start by plotting a zero-inflated poisson, since # of nest-searching queens is count data with a lot of zeros. I added bee_season_type as a covariate, since during my exploratory data analyses I saw that there was only a significant different in nest-searching queens among grassy vs bare sites during nesting season.

Steps

1. Try multiple random effects structures

- best fit = no random effect

```
NSQ_full <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                      bombus_floral_abun_scaled +
                      num_burrows_scaled +
                      bee_season_type +
                      (1|area/site_id/plot_id),
                    ziformula = ~ grass_forb_perc_scaled +
                      bombus_floral_abun_scaled +
                      num_burrows_scaled +
                      bee_season_type +
                      (1|area/site_id/plot_id),
                    family = poisson,
                    data = field_data)


NSQ_noArea <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                        bombus_floral_abun_scaled +
                        num_burrows_scaled +
                        bee_season_type +
                        (1|site_id/plot_id),
                      ziformula = ~ grass_forb_perc_scaled +
                        bombus_floral_abun_scaled +
                        num_burrows_scaled +
                        bee_season_type +
                        (1|site_id/plot_id),
                      family = poisson,
                      data = field_data)

NSQ_plotOnly <-  glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                        bombus_floral_abun_scaled +
                        num_burrows_scaled +
                        bee_season_type +
```

```r
                          (1|plot_id),
                        ziformula = ~ grass_forb_perc_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type +
                          (1|plot_id),
                        family = poisson,
                        data = field_data)

NSQ_siteOnly <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type +
                          (1|site_id),
                        ziformula = ~ grass_forb_perc_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type +
                          (1|site_id),
                        family = poisson,
                        data = field_data)

NSQ_AreaPlot <-  glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type +
                          (1|area/plot_id),
                        ziformula = ~ grass_forb_perc_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type +
                          (1|area/plot_id),
                        family = poisson,
                        data = field_data)

NSQ_noRE <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                      bombus_floral_abun_scaled +
                      num_burrows_scaled +
                      bee_season_type,
                    ziformula = ~ grass_forb_perc_scaled +
                      bombus_floral_abun_scaled +
                      num_burrows_scaled +
                      bee_season_type,  # zero-inflation model
                    family = poisson, #poisson
                    data = field_data)

#compare models using maximum likelihood test
anova(NSQ_full, NSQ_noArea, NSQ_plotOnly, NSQ_siteOnly, NSQ_AreaPlot, NSQ_noRE)
```

```
## Data: field_data
## Models:
## NSQ_noRE: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , zi=~grass_fo
## NSQ_noRE:     num_burrows_scaled + bee_season_type, zi=    bee_season_type, disp=~1
```

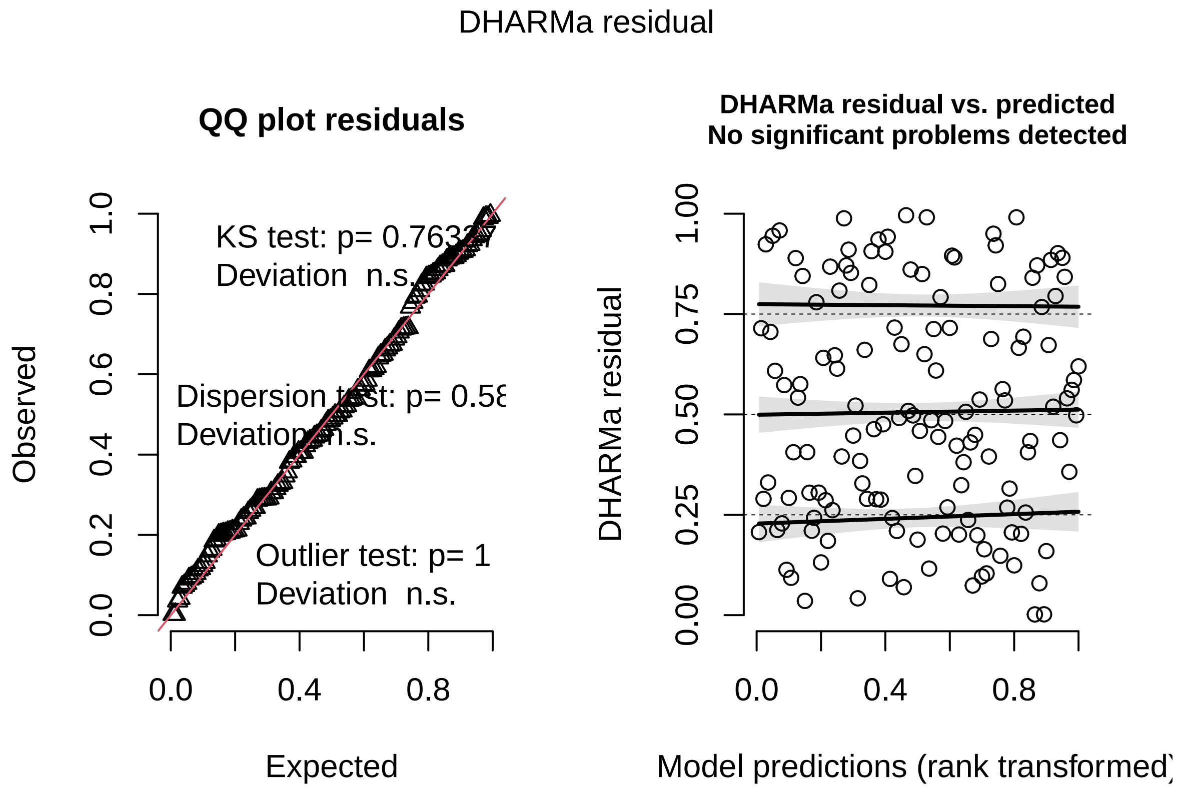
```
## NSQ_plotOnly: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , zi=~gras
## NSQ_plotOnly:     num_burrows_scaled + bee_season_type + (1 | plot_id), zi=    bee_season_type + (1
## NSQ_siteOnly: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , zi=~gras
## NSQ_siteOnly:     num_burrows_scaled + bee_season_type + (1 | site_id), zi=    bee_season_type + (1
## NSQ_noArea: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , zi=~grass_
## NSQ_noArea:     num_burrows_scaled + bee_season_type + (1 | site_id/plot_id), zi=    bee_season_type
## NSQ_AreaPlot: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , zi=~gras
## NSQ_AreaPlot:     num_burrows_scaled + bee_season_type + (1 | area/plot_id), zi=    bee_season_type +
## NSQ_full: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , zi=~grass_f
## NSQ_full:     num_burrows_scaled + bee_season_type + (1 | area/site_id/plot_id), zi=    bee_season_ty
##              Df    AIC    BIC  logLik deviance Chisq Chi Df Pr(>Chisq)
## NSQ_noRE     10 146.62 176.04 -63.313   126.62
## NSQ_plotOnly 12 150.62 185.93 -63.313   126.62     0      2          1
## NSQ_siteOnly 12 150.62 185.93 -63.313   126.62     0      0          1
## NSQ_noArea   14 154.62 195.81 -63.313   126.62     0      2          1
## NSQ_AreaPlot 14 154.62 195.81 -63.313   126.62     0      0          1
## NSQ_full     16 158.62 205.69 -63.313   126.62     0      2          1
```

```
##results tells us the model without random effects is best (lowest AIC).
```

2. Check diagnostics of best model.

- No significant problems detected.

```
residuals_nsq_noRE <- simulateResiduals(fittedModel =NSQ_noRE, plot = TRUE)
```



```r
##looks good
```

3. Look at results

```r
summary(NSQ_noRE)
```

```
##  Family: poisson  ( log )
## Formula:
## nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
##     num_burrows_scaled + bee_season_type
## Zero inflation:
## ~grass_forb_perc_scaled + bombus_floral_abun_scaled + num_burrows_scaled +
##     bee_season_type
## Data: field_data
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    146.6     176.0     -63.3     126.6       130
##
##
## Conditional model:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 0.07676    0.32263   0.238  0.81195
## grass_forb_perc_scaled     -0.00154    0.27836  -0.005  0.99559
## bombus_floral_abun_scaled  0.06287    0.24903   0.252  0.80068
## num_burrows_scaled          0.83361    0.29202   2.855  0.00431 **
## bee_season_typeworker      -0.30580    1.00347  -0.305  0.76057
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -0.04399    0.74875  -0.059   0.9531
## grass_forb_perc_scaled     -1.39254    0.65000  -2.142   0.0322 *
## bombus_floral_abun_scaled -0.57633    0.48987  -1.177   0.2394
## num_burrows_scaled          1.42124    0.67284   2.112   0.0347 *
## bee_season_typeworker       3.07732    1.37498   2.238   0.0252 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Findings

In conditional model (modelling abundance of nest-searching queens):

- no significant effect of grass+forb%, floral abundance, bee season
- significant positive effect of number of burrows

In zero-inflation model (modelling presence/absence of nest-searching queens):

- no significant effect of floral abundance

- Significant effect of grass+forb% (less likely to be absent with more grass/forb), number of burrows (more likely to be absent with more burrows), bee season (more likely to be absent during worker season than nest-searching season).

Interestingly, the number of rodent burrows is associated with more bombus nest-searching in the conditional model, but the number of rodent burrows is associated with more structural zeros in the zero-inflation model. This means that if a site is suitable for nesting, more rodent burrows is associated with more bombus nest-searching. However, in some cases, rodent burrows may signal inhospitable habitat conditions for bumble bees.

Number of workers vs nest-searching queens, floral abundance, rodent burrows.

Go through similar model selection steps as for the rodent burrow model.

Steps

1) Check different random effects structures

- best fit = 1|site_id/plot_id –> **area not included**

```
workers_full <- glmmTMB(workers ~ nest_searching_queens_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type +
                            (1|area/site_id/plot_id),
                        family = poisson,
                        data = field_data
                          )


workers_noArea <- glmmTMB(workers ~ nest_searching_queens_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type+
                            (1|site_id/plot_id),
                        family = poisson,
                        data = field_data
                          )

workers_plotOnly <-  glmmTMB(workers ~ nest_searching_queens_scaled +
                          bombus_floral_abun_scaled +
                          num_burrows_scaled +
                          bee_season_type+
                            (1|plot_id),
                        family = poisson,
                        data = field_data
                          )

workers_siteOnly <- glmmTMB(workers ~ nest_searching_queens_scaled +
```

```
                     bombus_floral_abun_scaled +
                     num_burrows_scaled +
                     bee_season_type+
                       (1|site_id),
                   family = poisson,
                   data = field_data
                     )

workers_AreaPlot <-  glmmTMB(workers ~ nest_searching_queens_scaled +
                     bombus_floral_abun_scaled +
                     num_burrows_scaled +
                     bee_season_type+
                       (1|area/plot_id),
                   family = poisson,
                   data = field_data
                     )

workers_noRE <- glmmTMB(workers ~ nest_searching_queens_scaled +
                     bombus_floral_abun_scaled +
                     num_burrows_scaled +
                     bee_season_type,
                   family = poisson,
                   data = field_data
                     )

#compare models using maximum likelihood test
anova(workers_full, workers_noArea, workers_plotOnly, workers_siteOnly, workers_AreaPlot, workers_noRE)
```

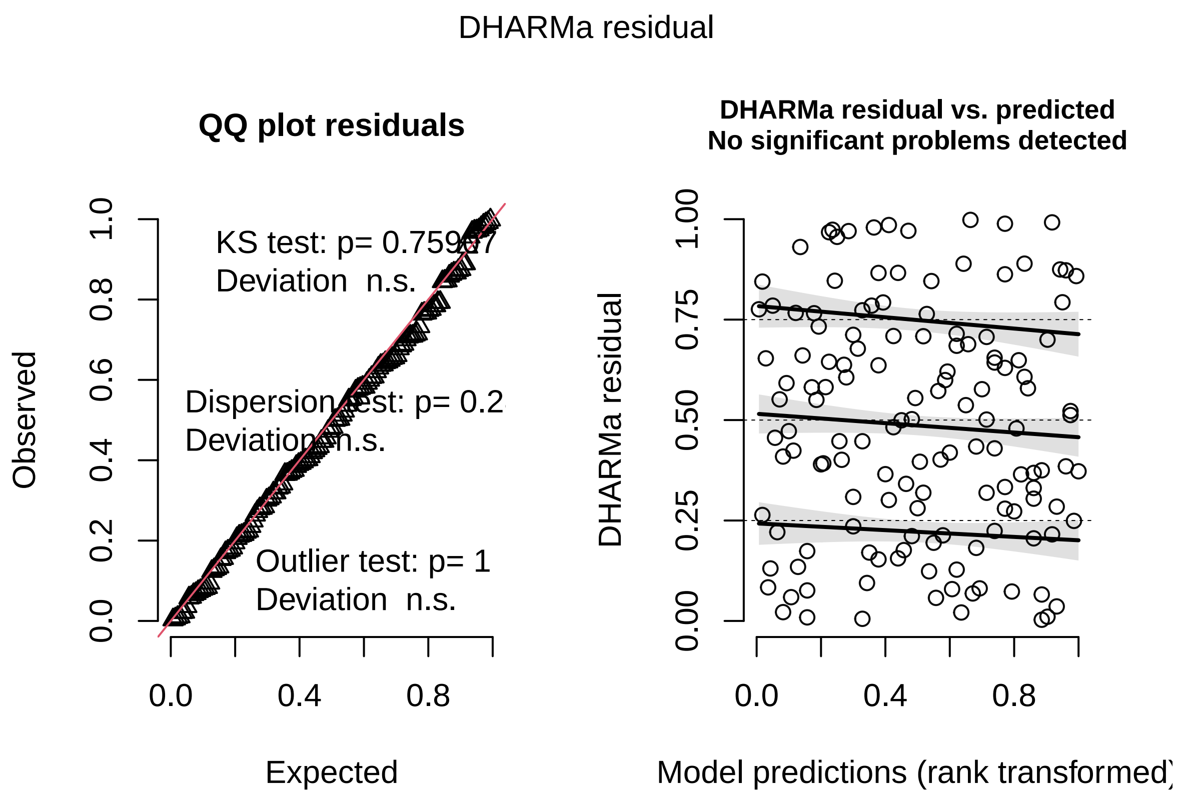
```
## Data: field_data
## Models:
## workers_noRE: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=~1
## workers_noRE:     num_burrows_scaled + bee_season_type, zi=~0, disp=~1
## workers_plotOnly: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=
## workers_plotOnly:     num_burrows_scaled + bee_season_type + (1 | plot_id), zi=~0, disp=~1
## workers_siteOnly: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=
## workers_siteOnly:     num_burrows_scaled + bee_season_type + (1 | site_id), zi=~0, disp=~1
## workers_noArea: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=~
## workers_noArea:     num_burrows_scaled + bee_season_type + (1 | site_id/plot_id), zi=~0, disp=~1
## workers_AreaPlot: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=
## workers_AreaPlot:     num_burrows_scaled + bee_season_type + (1 | area/plot_id), zi=~0, disp=~1
## workers_full: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=~1
## workers_full:     num_burrows_scaled + bee_season_type + (1 | area/site_id/plot_id), zi=~0, disp=~1
##                  Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
## workers_noRE      5 197.53 212.24 -93.767   187.53
## workers_plotOnly  6 196.94 214.59 -92.472   184.94 2.5891      1    0.10760
## workers_siteOnly  6 188.19 205.84 -88.097   176.19 8.7494      0    < 2e-16 ***
## workers_noArea    7 185.25 205.85 -85.627   171.25 4.9402      1    0.02624 *
## workers_AreaPlot  7 190.17 210.76 -88.085   176.17 0.0000      0    1.00000
## workers_full      8 187.25 210.79 -85.627   171.25 4.9152      1    0.02662 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##results tells us the model with no area is best (lowest AIC).
```

2. Check diagnostics of best model.

- Residual plots show that there are no significant issues with model fit. Check to see if negative binomial straightens lines in residual plot.

```
#simulate residuals
residuals_workers_noArea <- simulateResiduals(fittedModel = workers_noArea, plot = TRUE)
```

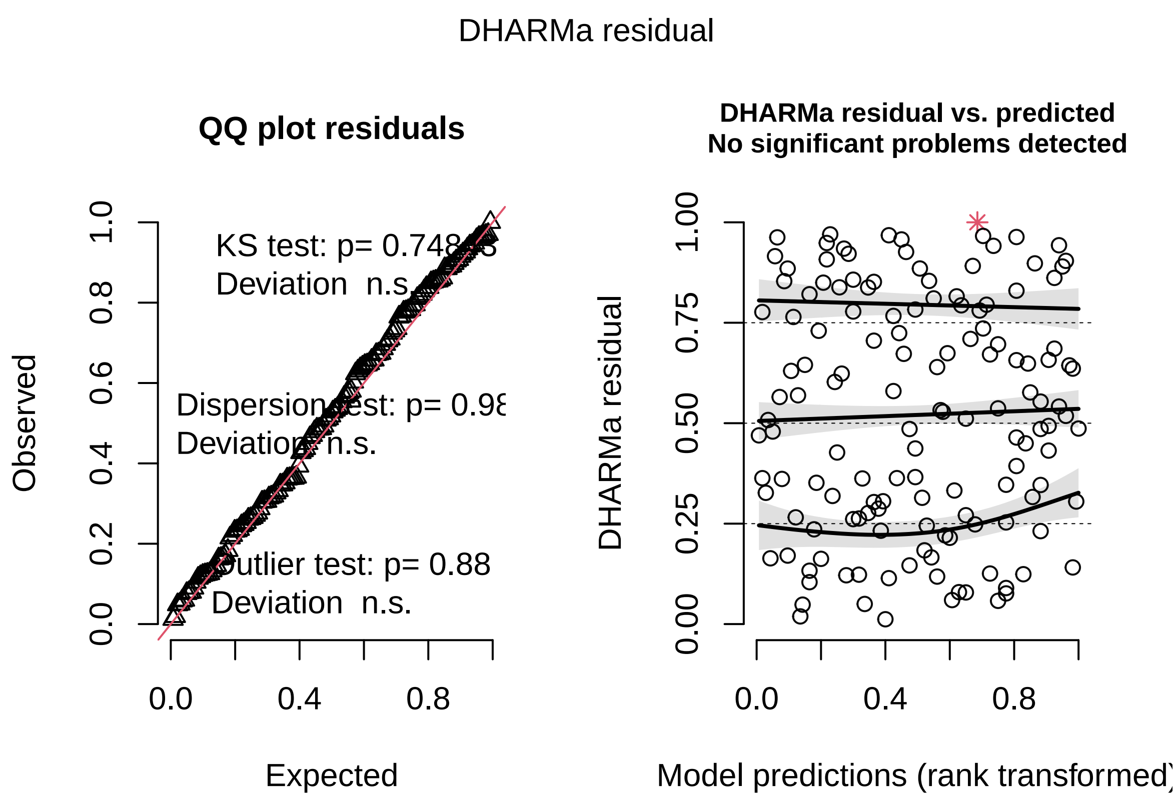


3. Plot negative binomial and check diagnostics.

- Residual plot shows better fit (straighter horizontal lines). AIC is also lower for negative binomial. This shows that the negative binomial is a better fit.

```
worker_negbin <- glmmTMB(workers ~ nest_searching_queens_scaled +
                       bombus_floral_abun_scaled +
                       num_burrows_scaled +
                       bee_season_type +
                         (1|site_id/plot_id),
                     family = nbinom2,
                     data = field_data
                       )

residuals_workers_negbin <- simulateResiduals(fittedModel = worker_negbin, plot = TRUE)
```




```
anova(workers_noArea, worker_negbin)
```

```
## Data: field_data
## Models:
## workers_noArea: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=~1
## workers_noArea:     num_burrows_scaled + bee_season_type + (1 | site_id/plot_id), zi=~0, disp=~1
## worker_negbin: workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled + , zi=~0, disp=~1
## worker_negbin:     num_burrows_scaled + bee_season_type + (1 | site_id/plot_id), zi=~0, disp=~1
##                Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
## workers_noArea  7 185.25 205.85 -85.627   171.25
## worker_negbin   8 161.67 185.20 -72.835   145.67 25.584      1  4.235e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##negative binomial is better (lower AIC)
```

4. Check results of final model

```
summary(worker_negbin)
```

```
##  Family: nbinom2  ( log )
## Formula:
## workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled +
##     num_burrows_scaled + bee_season_type + (1 | site_id/plot_id)
```

```
## Data: field_data
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     161.7     185.2     -72.8     145.7       132
##
## Random effects:
##
## Conditional model:
##  Groups          Name        Variance  Std.Dev.
##  plot_id:site_id (Intercept) 6.772e-12 2.602e-06
##  site_id         (Intercept) 4.216e-09 6.493e-05
## Number of obs: 140, groups:  plot_id:site_id, 24; site_id, 12
##
## Dispersion parameter for nbinom2 family (): 0.182
##
## Conditional model:
##                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                  -2.72109    0.56338  -4.830 1.37e-06 ***
## nest_searching_queens_scaled  0.07112    0.55093   0.129   0.8973
## bombus_floral_abun_scaled     0.67688    0.35602   1.901   0.0573 .
## num_burrows_scaled            0.08892    0.31759   0.280   0.7795
## bee_season_typeworker         1.79565    0.71167   2.523   0.0116 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Findings

Number of nest-searching queens and number of rodent burrows is does not significantly effect number of workers. Significantly more workers during worker season than during nest-searching season. Marginally insignificant (p=0.0573) positive effect of bombus-relevant floral abundance on worker abundance.

Final Path Diagram